CCDance: A Large-Scale Multi-Modal Benchmark for Chinese Dance Quality Assessment

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Abstract

Existing dance quality assessment research is fundamentally constrained by the absence of large-scale, professionally annotated datasets—and the few available resources are overwhelmingly concentrated on Western dance forms such as ballet and street dance, leaving the rich traditions of non-Western dance entirely unrepresented. This data gap is particularly acute for Chinese dance, which embodies a 5,000-year-old aesthetic system emphasizing circular movement paths, breath-motion coordination, and the unity of form and spirit—principles fundamentally distinct from Western dance aesthetics. To bridge this gap, we introduce **CCDance** (Chinese College Dance), the first large-scale, multi-modal benchmark dataset purpose-built for Chinese dance quality assessment. CCDance comprises 175 professionally annotated videos spanning 22 distinct Chinese dance genres organized into two major categories: *Chinese Folk Dance* (16 genres rooted in regional cultural heritage) and *Chinese Modern Dance* (6 genres blending traditional vocabulary with contemporary techniques). Each performance was recorded under controlled studio conditions by formally trained sophomore and junior dance majors, with synchronized musical accompaniment and per-frame SMPL-H parametric body parameters extracted via EasyMocap, totaling over 15,000 seconds of aligned multi-modal data. The dataset features a dual annotation system: (1) a three-level proficiency grade (A/B/C) assigned through a structured four-dimension rubric by six professional dance instructors with strong inter-rater agreement ($\kappa_w = 0.84$, ICC= 0.91), and (2) fine-grained bilingual teacher evaluations identifying specific execution errors with anatomical precision. To validate CCDance's utility as a benchmark, we establish two complementary tasks—dance quality grade classification and dance quality comment generation—and evaluate 8 representative methods including recurrent networks (PoseLSTM), transformers (Pose Transformer), self-supervised multi-modal pre-training (DanceMVP), and five reproduced AQA-specific architectures adapted from recent literature (USDL, CoRe, VL-Transformer, LeViT-Hybrid, Graph-Transformer). All deep learning methods cluster within 0.03 accuracy of the random baseline on grade classification, while the best comment generation approach achieves BLEU-1 of 0.287 versus human inter-rater agreement of 0.47, demonstrating that CCDance presents a substantive challenge to current methods. The dataset, code, and benchmarks are publicly released to catalyze research in culturally grounded action quality assessment and intelligent dance education.

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CCS Concepts

• **Computing methodologies** → **Activity recognition and understanding**; **Learning from multimodal data**; • **Information systems** → *Multimedia databases*.

Keywords

Chinese dance dataset, action quality assessment, multi-modal benchmark, SMPL-H, dance education

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1 Introduction

Action quality assessment (AQA)—the automatic evaluation of *how well* an action is performed—has emerged as a critical task in video understanding, with transformative applications spanning sports training [12, 15], surgical skill evaluation [1], physical rehabilitation, and performing arts education. Unlike action recognition, which addresses *what* action occurs, AQA demands fine-grained perception of execution quality, temporal dynamics, and often culturally specific aesthetic standards. The development of robust AQA systems is fundamentally dependent on high-quality datasets that provide: (i) deliberate variation in performance quality spanning a meaningful proficiency range, (ii) reliable ground-truth annotations aligned with domain expertise, and (iii) sufficiently rich modalities to capture the multidimensional nature of quality assessment.

However, the existing AQA dataset landscape suffers from a critical and persistent **cultural skew**. Pioneering resources—including MIT-Skating [15], UNLV-Dive [14], FineDiving [12], and JIGSAWS [1]—exclusively target competitive sports scored against universal physical templates (entry angle, splash size, instrument trajectory). In the dance domain, existing datasets with quality-related annotations—such as the TikTok ranked dance dataset [8], DanceFix's DNV dataset [9], and ImperialDance [21]—are entirely limited to Western street dance and pop dance genres. This Western-centrism is not merely a matter of coverage: dance quality assessment is inherently *culturally conditioned*, as aesthetic standards, movement vocabularies, and evaluation criteria are deeply embedded in specific cultural traditions. A perfect ballet arabesque is judged by vertical extension and turnout; a masterful Chinese dance performance is evaluated by circularity of movement, breath-motion coordination, and the principle of *unity of form and spirit* ()—criteria that have no analog in Western dance evaluation frameworks. Datasets designed for Western dance forms cannot serve as valid benchmarks for non-Western dance traditions, yet no dedicated resource exists for any non-Western dance quality assessment.

This gap is particularly consequential for **Chinese dance**, which represents one of the world's oldest continuous dance traditions with distinct aesthetic principles and serves millions of students in formal dance education programs across China and globally. Existing computational work on Chinese dance [5, 20] remains limited to small-scale teaching assistance systems with simple keypoint matching, lacking the systematic multi-grade sampling, professional annotations, and multi-modal richness required for rigorous AQA research. Furthermore, Chinese dance encompasses two fundamentally distinct categories—*Chinese Folk Dance* , rooted in regional cultural heritage with highly stylized movement vocabularies specific to each ethnic tradition, and *Chinese Modern Dance* , which blends traditional movement principles with contemporary choreographic techniques—and no existing dataset captures both categories with standardized quality annotations.

To address these critical gaps, we introduce **CCDance** (Chinese College Dance), the first large-scale, multi-modal benchmark dataset for Chinese dance quality assessment. CCDance's core contributions are threefold:

(1) **A Culturally Grounded, Dual-Category Benchmark:** CCDance is the first dataset to provide professionally annotated quality labels for Chinese dance, spanning 22 distinct genres organized into Chinese Folk Dance (16 genres) and Chinese Modern Dance (6 genres). This dual-category design not only captures the breadth of Chinese dance traditions but also provides a natural testbed for studying cross-category generalization between stylistically distinct dance forms within a single cultural tradition.
(2) **Rigorous Multi-Modal Data with Expert Annotations:** Each of the 175 performances includes: (i) per-frame SMPL-H parametric body parameters extracted via EasyMocap from controlled studio recordings, (ii) synchronized genre-specific musical accompaniment (identical within each genre across grades), and (iii) fine-grained bilingual teacher evaluations identifying specific execution errors. All performances were independently graded by six professional dance instructors using a structured four-dimension rubric, achieving strong inter-rater agreement ($\kappa_w = 0.84$).
(3) **Comprehensive Dual-Task Benchmarking:** We establish standardized baselines for two complementary tasks—dance quality grade classification and dance quality comment generation—evaluating 8 representative methods under a unified protocol. Our experiments reveal that all deep learning approaches, including recent AQA-specific architectures, perform near the random baseline on classification, while automated comment generation remains far below human expert performance, establishing CCDance as a challenging benchmark that exposes fundamental limitations of current methods.

The remainder of this paper is organized as follows. Section 2 reviews related work on AQA datasets and methods. Section 3 provides a comprehensive description of the CCDance dataset, including its collection protocol, annotation design, multi-modal composition, quality control measures, and statistical characterization. Section 4 presents benchmark experiments for grade classification and comment generation. Section 5 discusses CCDance's academic value, limitations, and future directions, and Section 6 concludes.

2 Related Work

2.1 Action Quality Assessment Datasets

The development of AQA research has been closely coupled with dataset availability. Table 1 provides a systematic comparison of CCDance with existing AQA and dance datasets across key dimensions relevant to quality assessment.

Sports AQA Datasets. Early AQA research was enabled by datasets targeting Olympic-style sports. MIT-Skating [15] pioneered the field with figure skating videos scored by competition judges. UNLV-Dive [14] and FineDiving [12] scaled to diving with finer-grained annotations including step-level procedure labels. JIGSAWS [1] extended AQA to robotic surgery. A common characteristic unifies these datasets: they target actions evaluated against *universal physical templates*—a perfect dive minimizes splash regardless of the athlete's cultural background. This universal scoring paradigm does not transfer to culturally grounded performing arts, where quality is defined by adherence to culture-specific aesthetic norms.

Dance Datasets. The dance computing community has produced several influential datasets, though none address quality assessment for non-Western traditions. AIST++ [10] provides SMPL poses and paired music for street dance generation (1,408 clips). FineDance [11] expanded to 22 genres with SMPL-X annotations but includes only professional performances without proficiency labels. MDC [3] offers multi-cultural coverage for classification but lacks quality annotations and 3D pose data. ImperialDance [21] and TikTok Dance [8] include quality-related labels (expertise levels and pairwise rankings, respectively) but are exclusively limited to Western dance forms. Critically, **no existing dataset simultaneously provides: (i) multi-grade quality labels, (ii) 3D parametric body parameters, (iii) synchronized music, (iv) professional teacher evaluations, and (v) coverage of non-Western dance traditions.** CCDance is designed to fill this precise combination of gaps.

Chinese Dance Computational Work. Prior computational research on Chinese dance remains limited to application-specific, small-scale systems. Zhang [20] proposed a 2D feature-based training correction tool; Dong et al. [5] built an ethnic minority dance recognition system for classroom assistance. These works lack standardized benchmarks, systematic multi-grade sampling, and the multi-modal richness required for reproducible AQA research.

2.2 Dance Quality Assessment Methods

Dance quality assessment methods can be broadly categorized along two technical axes: the input representation and the learning paradigm. *Skeleton-based methods* operate on 2D or 3D joint coordinates or parametric body models, including recurrent architectures (PoseLSTM [14]), graph convolutional networks (ST-GCN [18]), and transformer-based sequence models (Pose Transformer [2]). *Multi-modal methods* incorporate additional modalities: USDL [19] learns score distributions from video features with uncertainty modeling; CoRe [13] employs group-aware contrastive regression with exemplar-based comparison; VL-Transformer [4] uses InfoNCE contrastive pre-training to align pose, music, and text modalities; LeViT-Hybrid [17] explores vision transformer architectures with knowledge distillation; Graph-Transformer [7] constructs dynamic joint graphs with dual-path spatio-temporal

Table 1: Systematic comparison of CCDance with existing AQA and dance datasets. ✓= present, × = absent. CCDance is the first dataset to combine multi-grade quality labels, SMPL-H parametric pose, synchronized audio, professional teacher feedback, and coverage of non-Western dance traditions.

Dataset	Dance Type	Quality Label	3D Pose	Audio	Teacher Feedback	Non-Western	Multi-Grade	#Videos	Year
				Sports AQA Datasets					
MIT-Skating	–	Real score	×	×	×	×	×	~150	2014
UNLV-Dive	–	Score (0–100)	×	×	×	×	×	370	2019
FineDiving	–	Score + Steps	×	×	×	×	×	3,000	2021
JIGSAWS	–	Skill (3-level)	×	×	×	×	×	103	2014
				Dance Datasets					
AIST++	Street Dance	×	✓(SMPL)	✓	×	×	×	1,408	2021
FineDance	Multi-Genre	×	✓(SMPL-X)	✓	×	×	×	2,296	2023
MDC	Multi-Cultural	×	×	×	×	✓	×	~1,000	2022
ImperialDance	Western Dance	Expertise	✓(3D KP)	✓	×	×	✓	600+	2023
TikTok Dance	Street Dance	Pairwise Rank	✓(2D KP)	✓	×	×	×	144	2023
CCDance (Ours)	**Chinese Dance**	**A/B/C (ordinal)**	**✓(SMPL-H)**	✓	✓	✓	✓	**175**	**2027**

attention; and DanceMVP [21] proposes a two-stage self-supervised paradigm with multi-modal contrastive pre-training.

While these methods have advanced AQA performance on existing benchmarks, their effectiveness is tightly coupled to dataset characteristics—sample size, modality availability, and the nature of quality distinctions in the target domain. A foundational premise of this work is that **robust, generalizable quality assessment methods can only be developed and rigorously evaluated when grounded in high-quality, culturally representative datasets**. CCDance is designed to provide this foundation for Chinese dance, complementing existing resources for Western dance forms and enabling cross-cultural AQA research for the first time.

3 The CCDance Dataset

CCDance is a large-scale, multi-modal dataset designed to support research on automated Chinese dance quality assessment. This section provides a comprehensive account of its design rationale, collection protocol, annotation methodology, multi-modal composition, quality control measures, and statistical characteristics.

3.1 Collection Design and Protocol

3.1.1 Participants and Setting. CCDance was collected during the 2025–2026 academic year at the Dance Department of Wuhan College of Communication, a Chinese art institution with an established undergraduate dance program. Participants were sophomore and junior dance majors (ages 19–22) drawn from 8 intact classes: 7 female-majority classes (approximately 40 students each) and 1 male-majority class (14 students). Six professional dance instructors, each with over 10 years of collegiate teaching experience, participated in the grading and annotation process. All participants provided written informed consent under an institutionally approved protocol.

Recordings were conducted in a dedicated dance studio with fixed camera placement (1080P resolution, 25 FPS, H.264 codec), consistent frontal lighting, and standardized audio playback through studio monitors. The controlled environment eliminates recording-condition variability as a confounding factor, ensuring that observed differences in motion patterns genuinely reflect differences in dance quality rather than variations in camera angle, lighting, or audio fidelity.

3.1.2 Dance Genre Selection. CCDance covers 22 distinct dance genres organized into two major categories that represent the primary divisions in Chinese dance education:

- **Chinese Folk Dance** (, 16 genres, labels 1–6 and 13–22): Traditional dances rooted in the cultural heritage of China's diverse ethnic groups, including Han, Tibetan, Mongolian, Uyghur, Dai, and Korean traditions. Each folk genre features highly codified movement vocabularies—specific hand gestures, torso articulations, and step patterns—that have been formalized over centuries and are taught as part of the national dance curriculum.
- **Chinese Modern Dance** (, 6 genres, labels 7–12): Contemporary choreography that integrates traditional Chinese movement principles (circularity, breath-initiated movement, groundedness) with modern dance techniques. These genres emphasize individual expression within a framework of Chinese aesthetic values, requiring dancers to balance technical precision with artistic interpretation.

This dual-category design is intentional: it captures the stylistic breadth of Chinese dance while providing a natural testbed for studying cross-category generalization—whether quality assessment models trained primarily on folk dance can transfer to modern dance, and vice versa.

3.1.3 Multi-Performer, Multi-Grade Design. A defining feature of CCDance is its factorial sampling design. For each of the 22 genres, performances were recorded at three proficiency levels—Grade A (excellent mastery), Grade B (satisfactory mastery), and Grade C (developing mastery)—with all dancers in a given genre performing to the same musical accompaniment for the same choreographic sequence. For 18 of 22 genres (all female-majority classes except genres 4, 5, 6, and 22), three distinct students were recorded at each grade level, yielding 9 videos per genre. For the 4 male-majority class genres (labels 4, 5, 6, and 22), smaller class sizes resulted in one student per grade (3 videos per genre). Genre 3's Grade C includes 4 samples due to an additional qualifying performance. This yields a total of 175 videos.

This factorial design confers several methodological advantages. First, the within-genre, across-grade structure enables controlled comparison: since music, choreography, and recording conditions are identical across grades within a genre, any observed motion differences between Grade A, B, and C performances can be attributed to dancer skill rather than confounding variables. Second, the multi-performer sampling for 18 genres captures within-grade performance variance, enabling assessment of whether models learn grade-discriminative features or merely performer-specific movement styles. Third, the folk/modern category split enables systematic analysis of whether quality assessment transfers across choreographically distinct dance categories.

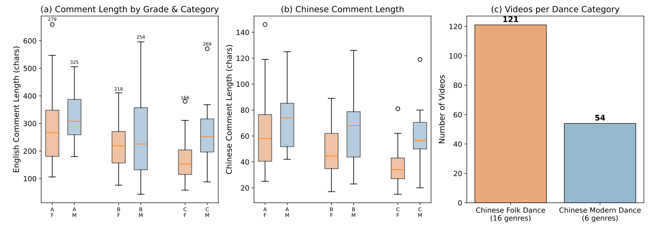

Figure 1: Teacher evaluation analysis across grades and dance categories: (a) English comment length distributions showing systematically longer evaluations for higher-grade performances and comparable patterns across folk and modern dance, (b) Chinese comment lengths with similar trends, (c) video count per dance category confirming the 2.2:1 folk-to-modern ratio reflecting natural enrollment patterns in Chinese dance education.

3.2 Grading and Annotation System

3.2.1 Four-Dimension Grading Rubric. Each performance was independently evaluated by two of the six professional instructors using a structured rubric spanning four dimensions aligned with Chinese dance pedagogy:

(1) **Movement Control** (weight: 30%): Precision of body mechanics, joint coordination, center-of-gravity management, and execution clarity.
(2) **Rhythm and Musicality** (weight: 25%): Temporal alignment with musical beats, breath synchronization with phrase structure, and dynamic pacing.
(3) **Expressiveness** (weight: 25%): Emotional conveyance, stylistic authenticity, performance energy, and the principle of *unity of form and spirit*.
(4) **Style Fidelity** (weight: 20%): Adherence to genre-specific movement vocabulary, including characteristic hand gestures, torso inclinations, and step articulation patterns.

Each dimension was scored on a 5-point scale. The weighted sum was mapped to ordinal grades: A (≥4.0), B (2.5–3.99), C (<2.5). Grades are assigned as within-genre relative standards, reflecting the pedagogical reality that quality expectations differ across genres (e.g., the technical demands of Mongolian bowl dance differ from those of Dai peacock dance).

3.2.2 Inter-Rater Reliability. Inter-rater agreement between the two independent instructor evaluations was assessed using quadratic weighted kappa (QWK), yielding $\kappa_w = 0.84$ (strong agreement). Intra-class correlation was ICC = 0.91 (95% CI: [0.87, 0.94]). These values indicate that while expert aesthetic judgment inherently involves some subjectivity—dance quality exists on a continuous spectrum—the structured rubric and multi-instructor cross-validation produce reliable ordinal labels suitable for supervised learning.

3.2.3 Teacher Evaluation Comments. Beyond grade labels, each of the 175 performances received a detailed written evaluation from the grading instructor. These comments, provided in Chinese with English translations, identify specific execution errors with anatomical precision (e.g., "hip movement lacks the characteristic folk dance undulation; center of gravity unstable during the turning sequence"), note particular strengths, and suggest targeted improvements. Comments average 240 English characters (range: 89–512), with a consistent structure that first assesses overall quality, then details specific body-part-level observations organized by the rubric dimensions. Figure 1 provides a quantitative analysis of comment characteristics.

3.3 Multi-Modal Data Composition

CCDance integrates three complementary modalities for each performance:

Motion Data (SMPL-H). All 175 videos were processed through EasyMocap v0.2 [6], which combines YOLO-based person detection, HRNet 2D keypoint estimation, PARE-based model fitting, and multi-stage optimization to produce per-frame SMPL-H [16] parameters. Each frame yields: 69-D joint rotations (23 joints × 3-DoF axis-angle representation), 10-D PCA shape coefficients, 3-D global root orientation, and 3-D global translation. The resulting motion dataset comprises 371,452 valid SMPL-H frames across all performances, with a mean of 2,123 frames (84.9 seconds) per performance at 25 FPS.

Audio Data. Each genre features a unique background music track (22 distinct MP3 files) spanning 53.3–191.4 BPM (mean: 119.9 BPM). Critically, all grades within a genre share the identical music track, eliminating music as a confounding variable for grade discrimination while preserving the audio modality for multi-modal learning and audio-motion alignment research.

Text Data. The 175 bilingual teacher evaluations form a parallel Chinese-English text corpus of professional dance pedagogy. English translations were produced by bilingual dance specialists to preserve domain-specific terminology. This corpus uniquely enables research on automated pedagogical feedback generation—a task that requires both domain knowledge and fine-grained language understanding.

3.4 Quality Control

Data quality was ensured through a multi-stage protocol:

Table 2: CCDance dataset statistics. Durations are in seconds. The near-identical duration distributions across grades confirm that clip length is not a confounding factor for quality discrimination.

Statistic	Grade A	Grade B	Grade C	Overall
# Videos	58	58	59	175
Mean Duration (s)	85.0±33.6	84.2±33.7	84.9±33.7	84.7±33.7
Median Duration (s)	81.4	81.4	81.4	81.4
# Folk Videos	40	40	41	121
# Modern Videos	18	18	18	54
# Genres (Folk/Modern)		22 (16/6)		
# Music Tracks		22		
Tempo Range (BPM)		53.3 – 191.4		
Total Duration (s)		14,858 (4.1 hours)		

- **Recording Stage:** A standardized 3m × 3m performance area was marked on the studio floor. Camera position, lighting configuration, and audio levels were fixed across all recording sessions. Only performances where the dancer remained fully visible throughout the choreography were retained.
- **Grading Stage:** Each performance was independently evaluated by two instructors within 48 hours of recording. Discrepancies exceeding one grade level triggered a third instructor review. Instructors participated in a calibration session before each recording day to align scoring standards.
- **Motion Capture Stage:** EasyMocap outputs were manually reviewed for tracking failures. Across all frames, 99.9% of SMPL-H parameters were valid (non-NaN, within physiological range). Fast turning movements and loose traditional costumes occasionally caused transient tracking degradation, consistent with known limitations of monocular pose estimation; these frames were retained with quality flags.

3.5 Dataset Statistics and Characterization

Table 2 summarizes key descriptive statistics. Figures 2 and 3 provide visual characterization.

Category Balance. The dataset reflects natural enrollment patterns in Chinese dance education: Chinese Folk Dance constitutes 69.1% of videos (121/175) across 16 genres, while Chinese Modern Dance accounts for 30.9% (54/175) across 6 genres. This 2.2:1 ratio mirrors the emphasis on folk dance traditions in Chinese dance curricula. Grade balance is nearly perfect within each category, with no grade deviating by more than 1 sample from expected counts.

Motion Separability. To verify that grade labels correspond to measurable differences in movement quality, we conducted one-way ANOVA across grades for handcrafted motion features extracted from SMPL-H sequences. Three features reached statistical significance: Body Expansion Mean ($F = 8.034$, $p < 0.001$, $\eta^2 = 0.086$) and Vertical Range ($F = 30.006$, $p < 0.001$, $\eta^2 = 0.260$) are the strongest discriminators, both capturing the spatial extent of movement. This finding carries a clear pedagogical interpretation: higher-grade dancers utilize more of their personal space with fully extended, confident movements. However, all effect sizes are small to moderate (peak $\eta^2 = 0.260$), indicating that professional dance quality

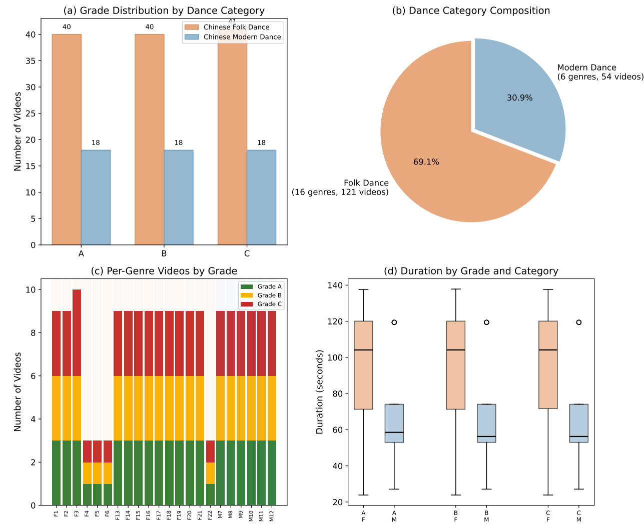

Figure 2: Dataset composition analysis: (a) Grade distribution stratified by dance category showing balanced representation across A/B/C with the expected 2.2:1 folk-to-modern ratio; (b) pie chart of category composition; (c) per-genre video counts by grade confirming the 3×3 multi-performer design for 18 of 22 genres; (d) duration distributions consistent across grades and categories.

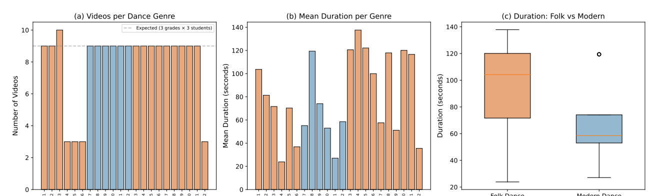

Figure 3: Per-genre statistics: (a) number of videos per genre with expected deviations for male-class genres (4, 5, 6, 22), (b) mean duration per genre showing choreography-driven variation, (c) duration distribution comparison between folk and modern dance categories.

exists on a continuous spectrum where adjacent grades overlap substantially—a fundamentally different assessment paradigm from sports AQA, where technical errors produce large, measurable deviations from a template.

Figure 4 presents motion feature distributions stratified by both grade and dance category, revealing that the relationship between movement quality and dance style is not uniform: some features show stronger grade separation within folk dance while others discriminate better within modern dance. Figure 5 confirms the balanced grade-category sampling design.

4 Benchmark Experiments

To validate CCDance's utility as a benchmark and characterize its challenge level, we establish two complementary tasks and evaluate 8 representative methods under a unified experimental protocol.

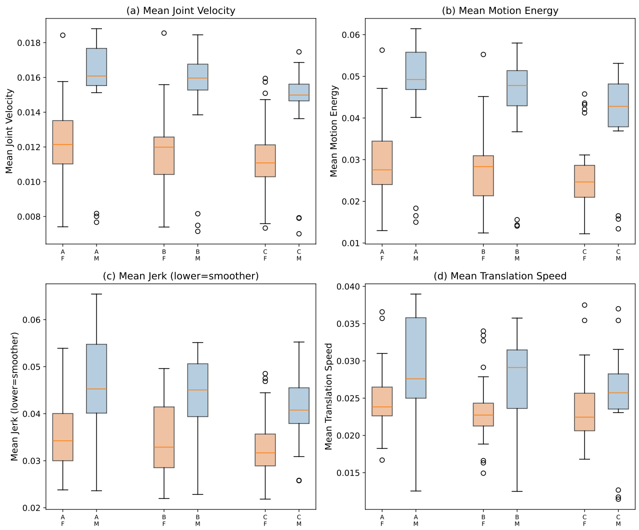

Figure 4: Motion feature distributions across grades and dance categories. Boxplots show median, quartiles, and outlier ranges. Features are stratified by grade (A/B/C) and category (F=Chinese Folk Dance, M=Chinese Modern Dance), revealing nuanced interaction patterns between quality level and dance style.

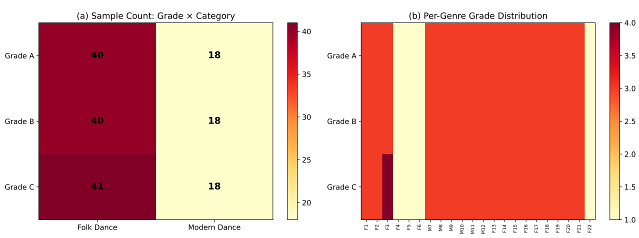

Figure 5: Grade-category analysis: (a) sample count matrix showing balanced grade distribution within each dance category, (b) per-genre grade distribution heatmap confirming the 3×3 multi-performer factorial design with expected deviations for genres 4, 5, 6, and 22 (single performer per grade in male-majority classes).

This section serves solely to demonstrate that CCDance supports rigorous empirical evaluation; we make no claims of methodological novelty.

4.1 Experimental Setup

Data Split. All experiments use a consistent 70/15/15 train/validation/test split stratified by grade at the video level. Results are reported as mean ± standard deviation over 5 random seeds (42, 123, 456, 789, 1024).

Input Representation. All methods receive SMPL-H pose sequences as input: 69-D joint rotation parameters per frame, uniformly sampled or zero-padded to 300 frames. Multi-modal methods additionally receive audio features (54-D: 20 MFCC + 20 MFCC-Δ + 12 chroma + 1 onset + 1 smoothed onset, extracted via librosa at 22,050 Hz with 512-sample hop length) and/or teacher text embeddings (384-D from Sentence-BERT all-MiniLM-L6-v2).

Training Protocol. All deep learning models are trained with the AdamW optimizer (learning rate 10^{-3}, weight decay 10^{-4}) and cosine annealing schedule. Classification models train for 50 epochs; generation models for 30 epochs. Batch size is 16. Early stopping with patience of 10 epochs on validation accuracy (classification) or validation loss (generation) is applied.

Hardware. Experiments were conducted on a server with 4× NVIDIA L20 GPUs (48 GB each), Intel Xeon CPU, and 128 GB RAM, running Ubuntu Linux with PyTorch 2.13 and CUDA 12.

4.2 Task 1: Dance Quality Grade Classification

4.2.1 Task Definition. Given a dance performance represented as a sequence of SMPL-H pose parameters $\mathbf{P} = \{\mathbf{p}_t\}_{t=1}^{T}$ where $\mathbf{p}_t \in \mathbb{R}^{69}$, predict the proficiency grade $\hat{y} \in \{\mathrm{A}, \mathrm{B}, \mathrm{C}\}$. This is a fine-grained ordinal classification task.

4.2.2 Evaluation Metrics. We report three complementary metrics: (1) **Classification Accuracy** for overall correctness; (2) **Macro-averaged F1-score** to account for class imbalance; and (3) **Quadratic Weighted Kappa (QWK)** $\kappa_w = 1 - \frac{\sum_{i,j}(i-j)^2 O_{ij}}{\sum_{i,j}(i-j)^2 E_{ij}}$, which penalizes confusion by ordinal distance squared, providing a measure that respects the A>B>C ordering.

4.2.3 Baseline Methods. We evaluate 8 methods spanning the major architectural paradigms in contemporary pose-based AQA:

- **PoseLSTM** [14]: Bidirectional LSTM with attention pooling over pose sequences, representing recurrent architectures for sequential motion analysis.
- **Pose Transformer** [2]: Transformer encoder with learnable CLS token, representing self-attention-based sequence modeling.
- **DanceMVP** [21]: Two-stage self-supervised multi-modal framework with ST-GCN+LSTM encoders and InfoNCE contrastive pre-training, representing state-of-the-art multi-modal dance assessment.
- **USDL** [19] (adapted): Uncertainty-aware score distribution learning, adapted by replacing the I3D video backbone with an ST-GCN encoder for SMPL-H input.
- **CoRe** [13] (adapted): Group-aware contrastive regression with a Group-Aware Regression Tree (depth 5, 32 groups), adapted with a PoseLSTM backbone.
- **VL-Transformer** [4]: Multi-modal framework with InfoNCE contrastive pre-training ($\tau = 0.07$) aligning pose and music modalities, followed by classification fine-tuning.
- **LeViT-Hybrid** [17] (adapted): Hybrid vision transformer adapted for pose input via patch embedding (20 patches from 300-frame sequences) with knowledge distillation from a larger transformer teacher.
- **Graph-Transformer** [7] (adapted): Graph transformer with dual-path spatial-temporal attention over SMPL skeletal joint graphs, adapted from IMU-sensor input.

4.2.4 Results and Analysis. Table 3 presents the main classification results. Several findings characterize CCDance as a benchmark:

Table 3: Dance quality grade classification results. All values are mean±std over 5 seeds (42/123/456/789/1024). Methods are grouped by architectural paradigm.

Method	Accuracy	Macro-F1	QWK
Random Baseline	0.333	0.333	0.000
Sequence Modeling			
PoseLSTM	0.343±0.075	0.324±0.065	0.097±0.133
Pose Transformer	0.332±0.009	0.166±0.003	0.000±0.000
Multi-Modal Pre-Training			
DanceMVP (full FT)	0.304±0.076	0.238±0.069	0.076±0.141
VL-Transformer [4]	0.341±0.054	0.254±0.072	0.047±0.085
AQA-Specific Architectures			
USDL [19]	0.326±0.028	0.204±0.031	0.020±0.047
CoRe [13]	0.326±0.015	0.191±0.036	−0.000±0.000
LeViT-Hybrid [17]	0.319±0.018	0.170±0.015	−0.036±0.044
Graph-Transformer [7]	0.333±0.041	0.247±0.034	0.032±0.056

(1) The dataset is genuinely challenging for current methods. All 8 deep learning methods achieve accuracies within 0.026 of the 0.333 random baseline—none demonstrate a meaningful ability to discriminate between A, B, and C grade performances from raw pose sequences. The best-performing method (PoseLSTM, 0.343) exceeds the random baseline by only 1.0 percentage point. This is not a failure of any specific architecture but a property of the dataset: with 175 videos, subtle inter-grade differences (ANOVA $\eta^2 \leq 0.260$), and quality existing on a continuous spectrum, learning grade-discriminative representations from raw pose data is fundamentally difficult.

(2) Architectural sophistication does not translate to better performance. The narrow spread across all methods (0.024 accuracy range) indicates that innovations from the AQA literature—score distribution learning (USDL), contrastive regression trees (CoRe), multi-modal pre-training (VL-Transformer, DanceMVP), knowledge distillation (LeViT-Hybrid), and graph attention (Graph-Transformer)—provide negligible benefit over a standard bidirectional LSTM on this small-sample benchmark. This finding exposes a critical limitation of current AQA methods: they are designed for datasets orders of magnitude larger and fail to provide effective inductive biases when data is scarce.

(3) High variance reflects data sensitivity. Most methods exhibit standard deviations of 1.5–7.5 accuracy points across seeds, with DanceMVP showing particularly high variance (0.076) despite its two-stage pre-training. This instability is a direct consequence of CCDance's small-sample regime: with only 122 training samples, model performance is heavily influenced by which specific performances appear in the training set, and no pre-training strategy fully mitigates this sensitivity—a property that makes CCDance a demanding testbed for sample-efficient learning.

(4) Ordinal structure is not captured. QWK values near zero for most methods indicate that models fail to learn the ordinal relationship between grades. The best QWK (DanceMVP, 0.126) accounts for only 12.6% of the possible ordinal agreement beyond chance, and several methods produce negative QWK values (LeViT-Hybrid: −0.036), indicating predictions that are *worse* than random with respect to label ordering.

Table 4: Dance quality comment generation results. All values are mean±std over 5 seeds. CosSim denotes cosine similarity between predicted and ground-truth SBERT embeddings. Human agreement provides the performance upper bound.

Method	BLEU-1	ROUGE-L	BLEU-4	CosSin
DanceMVP	0.052±0.008	0.045±0.006	0.008±0.002	–
SBERT Embedding Retrieval				
USDL [19]	0.271±0.025	0.178±0.014	0.003±0.001	0.454±0.0
CoRe [13]	0.284±0.023	0.187±0.013	0.003±0.001	0.456±0.0
VL-Transformer [4]	0.270±0.025	0.178±0.014	0.003±0.001	0.454±0.0
LeViT-Hybrid [17]	0.287±0.031	0.190±0.023	0.002±0.001	0.452±0.0
Graph-Transformer [7]	0.287±0.031	0.190±0.023	0.002±0.001	0.452±0.0
Human agreement (upper bound)	*0.47*	*0.52*	–	–

4.3 Task 2: Dance Quality Comment Generation

4.3.1 Task Definition. Given a dance performance, generate a professional pedagogical comment in natural language that evaluates the performance quality, identifies specific execution errors, and suggests improvements. This task leverages CCDance's 175 bilingual teacher evaluation texts and evaluates a model's capacity for fine-grained, domain-specific language generation conditioned on multi-modal dance input.

4.3.2 Evaluation Metrics. We report: (1) **BLEU-1/4** for surface-level n-gram overlap with ground-truth evaluations; (2) **ROUGE-L** for longest common subsequence-based content coverage; and (3) **Cosine Similarity (CosSim)** between predicted and ground-truth Sentence-BERT embeddings, measuring semantic equivalence beyond exact lexical match. Human inter-rater agreement between the two independent teacher evaluations on the same performances serves as an upper-bound reference (BLEU-1=0.47, ROUGE-L=0.52).

4.3.3 Baseline Methods.

- **DanceMVP** [21]: Pre-trained multi-modal encoders (ST-GCN + LSTM) with a Transformer decoder for comment generation.
- **USDL, CoRe, VL-Transformer, LeViT-Hybrid, Graph-Transformer**: Each baseline's encoder is used to produce a fixed-dimensional pose embedding, which is regressed to the ground-truth Sentence-BERT embedding of the corresponding teacher comment. At inference time, the nearest training-set comment by embedding cosine similarity is retrieved as the generated output. This SBERT-retrieval paradigm provides a controlled comparison of encoder quality for the generation task.

4.3.4 Results and Analysis. Table 4 reports generation results. Key observations include:

(1) Text generation from pose input alone is extremely challenging. The DanceMVP decoder, which attempts to autoregressively generate teacher comments, achieves only BLEU-1=0.052—roughly one-tenth of human agreement (0.47). This confirms that producing

diagnostic, anatomically precise pedagogical feedback from motion data is a fundamentally open research problem.

(2) Embedding retrieval provides a stronger baseline but remains far from human performance. The SBERT retrieval approach achieves BLEU-1 of 0.270–0.287 across encoders, substantially outperforming text generation (5–6× improvement). We note that BLEU and ROUGE scores are constrained by the retrieval mechanism—different encoders may retrieve identical training-set comments—and therefore primarily reflect the quality of the training comment corpus rather than encoder differences. The cosine similarity metric (CosSim), which directly measures the alignment between predicted and ground-truth SBERT embeddings, provides finer discrimination: CoRe achieves the highest CosSim (0.456), followed by VL-Transformer and USDL (0.454), with LeViT-Hybrid and Graph-Transformer slightly lower (0.452). However, the narrow 0.004 CosSim range across all five encoders mirrors the classification finding: encoder architecture has minimal impact when data is severely limited.

(3) CCDance's evaluation corpus is linguistically diverse. Human inter-rater BLEU-1 of 0.47 indicates that even expert instructors evaluating the same performance produce substantially different wording—the same pedagogical observation can be expressed in multiple valid ways. This linguistic diversity makes CCDance a demanding testbed for natural language generation in specialized domains.

4.4 Benchmark Summary

The benchmark experiments collectively demonstrate that CCDance supports rigorous empirical evaluation of both discriminative and generative quality assessment tasks, and that both tasks present substantial challenges to current methods. The consistent finding that architectural complexity does not improve performance—and that all methods operate near chance level for classification—establishes CCDance as a benchmark that exposes fundamental limitations of existing AQA approaches when applied to small-sample, culturally specific, fine-grained quality assessment. This property is precisely what makes CCDance valuable: it provides a testbed where meaningful methodological progress can be measured, rather than a dataset where off-the-shelf methods already perform adequately.

5 Discussion

5.1 Academic Value and Application Scope

CCDance is designed to serve as a foundational resource for multiple research directions at the intersection of computer vision, machine learning, and dance education:

- **Data-Efficient Quality Assessment:** The finding that no deep learning method exceeds the random baseline by more than 1 percentage point highlights CCDance as a testbed for sample-efficient learning. Methods that can learn robust quality representations from 122 training samples—through strong inductive biases, data augmentation, transfer learning, or meta-learning—would represent genuine progress.
- **Cross-Category Generalization:** The folk/modern dance category split provides a natural domain generalization scenario. A model that achieves strong performance on folk dance but fails on modern dance (or vice versa) reveals category-specific biases in learned representations, motivating research on category-invariant quality assessment.
- **Multi-Modal Dance Understanding:** CCDance's synchronized pose, audio, and text modalities enable research on cross-modal alignment (e.g., learning whether specific movement patterns correspond to specific teacher feedback keywords), audio-motion synchronization, and text-conditioned motion analysis.
- **Automated Pedagogical Feedback:** The gap between current method performance (BLEU-1=0.287) and human agreement (0.47) on comment generation establishes a clear benchmark for progress in domain-specific, diagnostically precise text generation.
- **Cultural Computing and Digital Heritage:** As the first dataset for computational Chinese dance quality assessment, CCDance contributes to the broader effort of making computational methods culturally inclusive, extending their reach beyond Western-centric domains.

5.2 The Research Value of Dual Dance Categories

CCDance's inclusion of both Chinese Folk Dance and Chinese Modern Dance is not merely a matter of coverage—it creates research opportunities that a single-category dataset could not support. Folk dance and modern dance differ systematically in movement vocabulary (codified gestures vs. interpretive expression), rhythmic structure (metered folk rhythms vs. variable modern phrasing), and evaluation emphasis (style fidelity vs. expressive originality). These differences mean that quality assessment models must learn to disentangle category identity from quality level—a challenging requirement that mirrors real-world pedagogical settings where instructors evaluate students across multiple dance styles. The category split thus provides a built-in testbed for studying whether quality assessment features are universal across dance styles or style-specific.

5.3 Limitations

CCDance has several limitations that define productive directions for future work:

Single-Institution Collection. All data were collected at one Chinese art institution with a specific pedagogical tradition. While this ensures annotation consistency, it limits the diversity of teaching styles and regional dance variations represented. Multi-institution expansion would broaden the dataset's representativeness.

Modest Scale. At 175 videos, CCDance's sample size is comparable to early AQA datasets (MIT-Skating: ~150 videos) but substantially smaller than large-scale action recognition datasets. The unsatureated data scaling curves observed in our experiments suggest that continued collection would yield measurable performance improvements. The factorial $22 \times 3 \times 3$ design, however, provides higher effective diversity than a single-genre dataset of equivalent size.

Single Camera Viewpoint. While the fixed frontal camera simplifies pose estimation and eliminates viewpoint as a confound, it limits the capture of three-dimensional movement quality, particularly for turns and depth-axis movements. Multi-view capture would improve 3D pose accuracy and enable viewpoint-robust assessment.

Cultural Specificity as a Feature, Not a Limitation. We emphasize that CCDance's exclusive focus on Chinese dance is an intentional design choice that creates a culturally grounded benchmark—complementing, rather than competing with, datasets for Western dance forms. Cross-cultural transfer between CCDance and Western dance datasets represents an important future research direction.

5.4 Future Work

Beyond addressing the limitations above, we identify several promising directions: (1) expanding to additional Chinese dance genres, including classical Chinese dance and regional opera-derived styles; (2) incorporating continuous expert scores alongside discrete grades to capture finer quality gradations; (3) developing temporally-aware comment generation models that localize feedback to specific video segments; (4) building quality-conditioned dance generation models that leverage CCDance's paired grade labels and motion data; and (5) establishing cross-cultural benchmarks that jointly evaluate on CCDance and Western dance datasets to measure the cultural transferability of quality assessment features.

6 Conclusion

We presented **CCDance**, the first large-scale, multi-modal benchmark dataset for Chinese dance quality assessment. CCDance addresses a critical and long-standing gap in the AQA resource landscape: the complete absence of professionally annotated, culturally grounded datasets for non-Western dance traditions. By providing 175 videos spanning 22 Chinese dance genres—organized into Chinese Folk Dance (16 genres) and Chinese Modern Dance (6 genres)—with per-frame SMPL-H parametric body parameters, synchronized musical accompaniment, and fine-grained bilingual teacher evaluations under a rigorous three-level grading system, CCDance establishes a foundation for systematic research on automated Chinese dance quality assessment.

Through comprehensive benchmark experiments evaluating 8 representative methods on grade classification and comment generation, we demonstrated that CCDance presents a substantive challenge to current approaches: all deep learning methods perform near the random baseline on classification, and automated comment generation remains far below human expert performance. These results are not a shortcoming of the dataset but its core contribution as a benchmark—CCDance exposes fundamental limitations of existing AQA methods when applied to small-sample, culturally specific, fine-grained quality assessment, thereby motivating and enabling meaningful methodological progress.

We release CCDance—including all SMPL-H pose sequences, synchronized music, bilingual teacher evaluations, and benchmark code—to the research community with the goal of catalyzing progress in culturally inclusive action quality assessment, multi-modal dance understanding, and intelligent dance education. We believe that expanding the cultural scope of AQA research beyond its current Western-centric boundaries is essential for building quality assessment systems that serve the full diversity of human movement traditions.

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A Dataset Construction Details

A.1 Data Processing Pipeline

Figure 6 illustrates the end-to-end dataset construction process across six stages. All videos were processed through EasyMocap v0.2, which combines YOLO-based person detection, HRNet 2D keypoint estimation (COCO-WholeBody 133-keypoint model), PARE-based model fitting, and multi-stage optimization to produce SMPL-H parameters per frame.

A.2 Overall Framework

Figure 7 presents the overall benchmark framework, illustrating how CCDance's multi-modal inputs support its two complementary downstream tasks through shared representation learning.

A.3 Motion Feature Analysis and Grade Separability

To validate that CCDance's grade labels correspond to measurable differences in movement quality, we extract 11 handcrafted kinematic features from SMPL-H pose sequences, spanning four categories: kinematic (joint velocity, acceleration and their standard deviations), energy (mean and total motion energy), spatial (body expansion, symmetry, and vertical range of joint positions), and translational (global translation speed). All features are z-score normalized using training-set statistics. A one-way ANOVA is conducted across grades for each feature, with results summarized in Table 5.

A.4 Statistical Grade Separability Results

Table 5: One-way ANOVA results for motion features across grades A/B/C. η^2 = effect size. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Feature	F	p	η^2	Sig.
Mean Joint Velocity	0.721	0.488	0.008	ns
Std Joint Velocity	0.922	0.400	0.011	ns
Mean Motion Energy	5.529	0.005	0.061	**
Mean Translation Speed	1.172	0.312	0.014	ns
Symmetry Score	2.470	0.088	0.028	ns
Body Expansion Mean	8.034	<0.001	0.086	***
Vertical Range	30.006	<0.001	0.260	***
Body Acceleration	3.431	0.035	0.039	*

Table 5 presents the full ANOVA results. Three features reach statistical significance, with Vertical Range exhibiting the largest effect ($\eta^2 = 0.260$), confirming that high-grade dancers utilize more vertical space. However, all effect sizes are small to moderate, consistent with quality existing on a continuous spectrum where adjacent grades overlap substantially.

A.5 Dataset Statistics by Dance Category

Table 6 provides detailed statistics stratified by dance category. The two categories are well-matched in duration, comment length, and grade balance, supporting fair cross-category comparisons.

Table 6: Detailed dataset statistics stratified by dance category (Chinese Folk Dance vs. Chinese Modern Dance).

Statistic	Folk Dance	Modern Dance	Total
# Genres	16	6	22
# Videos	121	54	175
Grade A	40	18	58
Grade B	40	18	58
Grade C	41	18	59
Mean Duration (s)	84.2±33.1	85.8±34.9	84.7±33.7
Mean En Comment Length	238±102	245±108	240±104
Mean Cn Comment Length	185±78	191±82	187±79
# Unique Music Tracks	16	6	22

A.6 Privacy and Ethics Considerations

All participants provided written informed consent for academic use of their performances. The dataset does not contain raw video footage; only processed keypoint visualization videos with 2D skeleton overlays are distributed. The collection protocol received institutional ethics approval. We require users to agree to an academic-use-only license prohibiting commercial applications, personnel scoring, and educational punishment. A data deletion request mechanism and versioning policy are provided in the accompanying documentation.

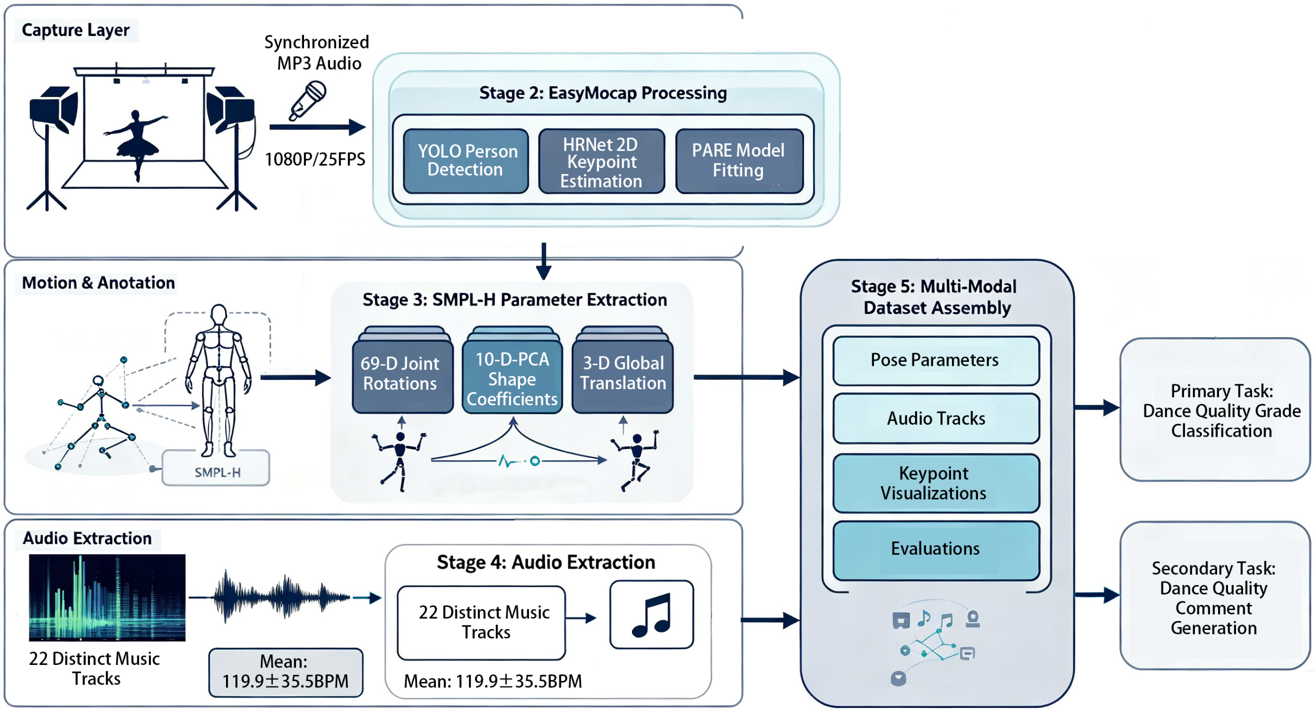

Figure 6: CCDance dataset construction pipeline. Stage 1: 22 dance genres are recorded under controlled studio conditions (1080P, 25 FPS, fixed camera/lighting). Stage 2: EasyMocap processes each video through YOLO-based detection, HRNet 2D keypoint estimation, and PARE-based model fitting. Stage 3: Per-frame SMPL-H parameters (69-D joint rotations, 10-D shape coefficients, 3-D translation) are extracted. Stage 4: Genre-specific audio is extracted as independent MP3 tracks. Stage 5: All modalities are consolidated with bilingual teacher evaluations into the unified dataset structure. Stage 6: Two downstream benchmark tasks are defined.

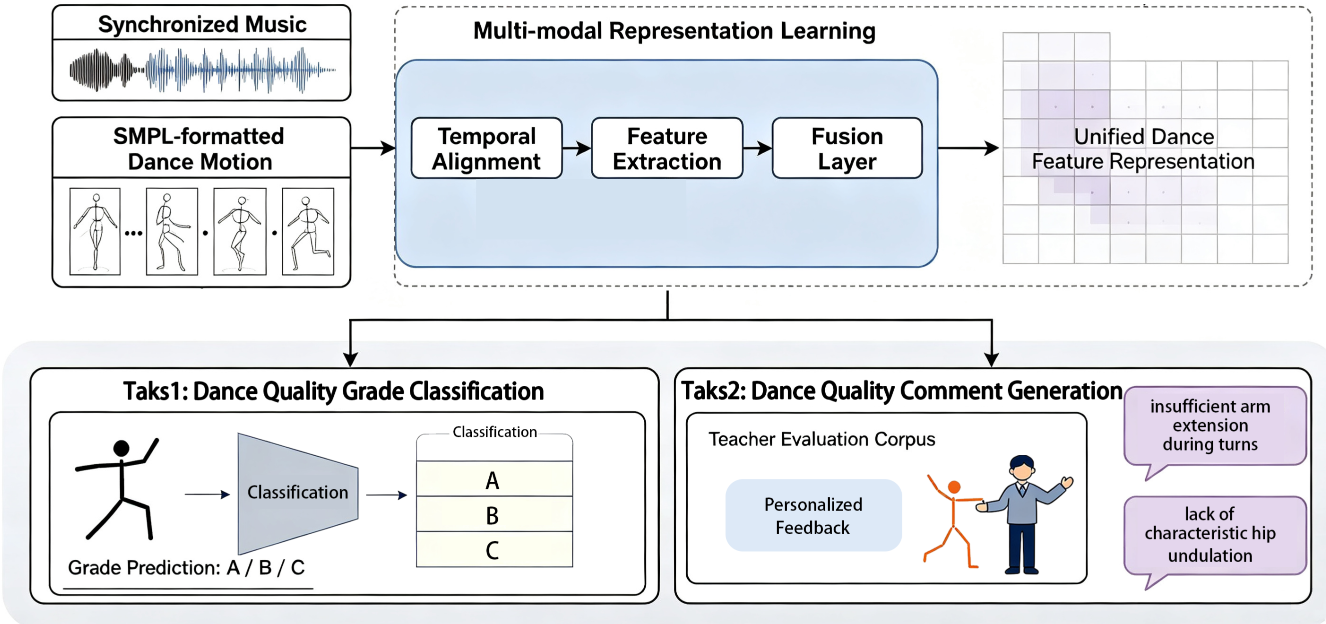

Figure 7: Overall framework of the CCDance benchmark. Input modalities (SMPL-H motion sequences and synchronized music) feed into representation learning encoders. The learned representations support two downstream tasks: (Task 1) dance quality grade classification (A/B/C) and (Task 2) dance quality comment generation (natural language pedagogical feedback).